THE DUALIZING COMPLEX OF A PROPER SCHEME OVER A FIELD IS INDEPENDENT OF THE FIELD OF DEFINITION AND COMPATIBLE WITH FIELD EXTENSIONS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FW1 (sources/stacks-duality/pinned-original-source.tex line 7163).

Allowed standard prerequisites (editorial).

- The dualizing complex of a proper scheme over a field and the description of the adjoint on affines. Same chapter, Lemma labelled lemma-duality-proper-over-field and Example labelled example-affine-twisted-inverse-image, cited at proof lines 7207-7219 and 7229-7237. Fixes the objects and the affine computation of the shriek of a field.
- Finiteness of the field extension arising from the factorisation. Cohomology of Schemes, Lemma labelled coherent-lemma-proper-over-affine-cohomology-finite, cited at proof lines 7221-7223. Makes k'/k finite.
- Base change for the right adjoint along a flat morphism. Same chapter, Lemma labelled lemma-more-base-change and Remark labelled remark-relative-dualizing-complex, cited at proof lines 7239-7240. Gives part (2).

Source reference. Original lemma 0FVV: sources/stacks-duality/pinned-original-source.tex, source lines 6934–6934. Same-chapter pointer into the author environment lemma at source line 6934; invoked at proof line 7181 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line 7184 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 02O6: sources/stacks-chow/tags.txt, source lines 3505–3505. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7191 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0A9F: sources/stacks-duality/pinned-original-source.tex, source lines 681–681. Same-chapter pointer into the author environment example at source line 681; invoked at proof line 7197 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA8: sources/stacks-duality/pinned-original-source.tex, source lines 1636–1636. Same-chapter pointer into the author environment lemma at source line 1636; invoked at proof line 7203 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0B6S: sources/stacks-duality/pinned-original-source.tex, source lines 2945–2945. Same-chapter pointer into the author environment remark at source line 2945; invoked at proof line 7204 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a proper scheme over a field k. Let $\omega_X^\bullet$ and ω_X be as in Lemma 0FVV.*

(1) *If $X \to \mathrm{Spec}(k)$ factors as $X \to \mathrm{Spec}(k') \to \mathrm{Spec}(k)$ for some field k', then $\omega_X^\bullet$ and ω_X are as in Lemma 0FVV for the morphism $X \to \mathrm{Spec}(k')$.*
(2) *If K/k is a field extension, then the pullback of $\omega_X^\bullet$ and ω_X to the base change X_K are as in Lemma 0FVV for the morphism $X_K \to \mathrm{Spec}(K)$.*

Proof. Denote $f : X \to \mathrm{Spec}(k)$ the structure morphism and denote $f' : X \to \mathrm{Spec}(k')$ the given factorization. In the proof of Lemma 0FVV we took $\omega_X^\bullet = a(\mathcal{O}_{\mathrm{Spec}(k)})$ where a be is the right adjoint of Lemma 0A9E for f. Thus we have to show $a(\mathcal{O}_{\mathrm{Spec}(k)}) \cong a'(\mathcal{O}_{\mathrm{Spec}(k)})$ where a' be is the right adjoint of Lemma 0A9E for f'. Since $k' \subset H^0(X, \mathcal{O}_X)$ we see that k'/k is a finite extension (Cohomology of Schemes, Lemma 02O6). By uniqueness of adjoints we have $a = a' \circ b$ where b is the right adjoint of Lemma 0A9E for $g : \mathrm{Spec}(k') \to \mathrm{Spec}(k)$. Another way to say this: we have $f^! = (f')^! \circ g^!$. Thus it suffices to show that $\mathrm{Hom}_k(k', k) \cong k'$ as k'-modules, see Example 0A9F. This holds because these are k'-vector spaces of the same dimension (namely dimension 1).

Proof of (2). This holds because we have base change for a by Lemma 0AA8. See discussion in Remark 0B6S. □